

ORGNZ-05: Bucket-Level Access Control

Series: ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 5 of 10 | **Created:** January 2026 | **Last Updated:** 05/06/2026

Overview

Bucket-level access control provides a straightforward way to isolate data by team, application, or business unit. By granting permissions to specific buckets, you can ensure teams only access data relevant to their responsibilities.

Canonical pattern (when bucket-level access fits): Bucket-level access works well in specific scenarios — **compliance separation** (PCI/HIPAA), **retention isolation** (different retention requirements naturally produce distinct buckets), **hard cost attribution** (per-LOB billing buckets), or **hostile multi-tenancy** (strict cross-tenant isolation). For general team-scoped data access without one of those scenarios, `dt.security_context` + **record-level permissions** (covered in **ORGNZ-06**) is usually the simpler choice — it scales beyond the 80-bucket limit and is mutable without re-routing ingest. See [IAM REFERENCE.md § Bucket-Match Overlay](#) for the canonical scenario gating. The patterns below teach bucket-level mechanics for when the scenario fits.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	<code>storage:bucket-definitions:read</code> , <code>storage:logs:read</code>
Knowledge	Completed ORGNZ-04 (Permissions in Grail)

Requirement	Details
Data	At least 1 hour of log data

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Learning Objectives

By the end of this notebook, you will:

- Create IAM policies for bucket-level access
- Use bucket naming patterns in policies
- Implement team isolation using buckets
- Understand bucket permission best practices

Bucket Permission Fundamentals

Required Permissions

All bucket access policies must start with `storage:buckets:read` :

```
ALLOW storage:buckets:read WHERE <condition>;  
ALLOW storage:logs:read; // Then allow table access
```

Two-Step Pattern

Step	Purpose	Example
1. Bucket access	Define which buckets	<code>storage:buckets:read WHERE bucket-name = 'x'</code>
2. Table access	Define which data types	<code>storage:logs:read , storage:metrics:read</code>

Policy Examples

Example 1: Single Bucket Access

Grant a team access to their specific bucket:

```
{
  "name": "platform-team-logs-access",
  "description": "Platform team can access platform logs bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name = 'team_platform_logs'; ALLOW storage:logs:read;",
  "tags": ["team:platform"]
}
```

Example 2: Multiple Buckets with IN Operator

Grant access to a list of specific buckets:

```
{
  "name": "finance-multi-bucket-access",
  "description": "Finance team can access multiple buckets",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name IN ('finance_logs', 'finance_metrics', 'finance_audit'); ALLOW storage:logs:read, storage:metrics:read;",
  "tags": ["team:finance"]
}
```

Example 3: Bucket Name Pattern with STARTSWITH

Grant access to all buckets matching a prefix:

```
{
  "name": "prod-infrastructure-access",
  "description": "Access to all production infrastructure buckets",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'prod_infra_'; ALLOW storage:logs:read, storage:metrics:read,
storage:spans:read;",
  "tags": ["env:production"]
}
```

Example 4: Bucket Pattern with MATCH

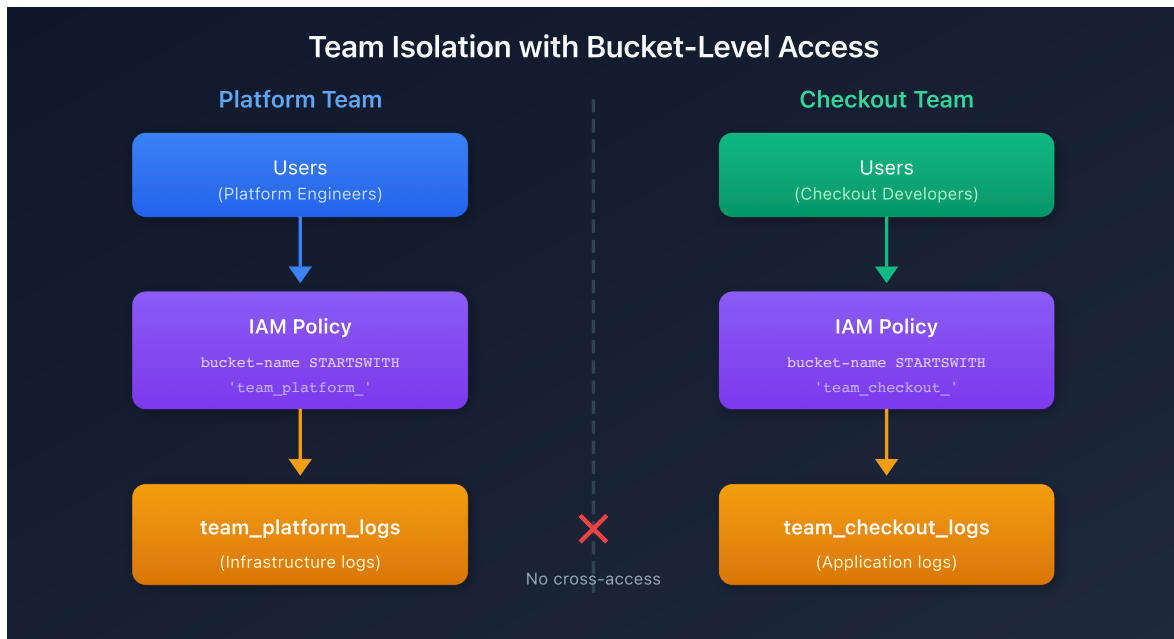
Use pattern matching for complex bucket names:

```
{
  "name": "database-team-access",
  "description": "Database team can access any database-related bucket",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
MATCH ('*-database-*'); ALLOW storage:logs:read;",
  "tags": ["team:database"]
}
```

Team Isolation Pattern

When this approach fits: team isolation via dedicated buckets is well-suited to scenarios where each team's data already needs its own bucket for compliance, retention, or hard-cost-attribution reasons. For routine per-team scoping without one of those reasons, `dt.security_context` **parameterization** (one policy template, bound per team — see **ORGNZ-06** and **IAM-04 Pattern 2**) is usually simpler and avoids the 80-bucket limit. The pattern below applies once you've decided buckets are the right tool for your team-isolation scenario.

Architecture



Implementation

Step 1: Create team buckets

```
team_platform_logs
team_checkout_logs
team_payments_logs
```

Step 2: Route data via OpenPipeline

```
processors:
- type: route
  rules:
    - condition: "host.group starts-with 'platform-'"
      destination: "team_platform_logs"
    - condition: "service.name contains 'checkout'"
      destination: "team_checkout_logs"
```

Step 3: Create IAM policies per team

```
// Platform team policy
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH
"team_platform_";
ALLOW storage:logs:read;

// Checkout team policy
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH
"team_checkout_";
ALLOW storage:logs:read;
```

Default Buckets Policy

Standard Default Access

For users who need access to all default buckets:

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:events:read, storage:logs:read, storage:metrics:read,
      storage:entities:read, storage:bizevents:read, storage:spans:read;
```

Default Plus Custom Buckets

```
ALLOW storage:buckets:read WHERE storage:bucket-name STARTSWITH "default_";
ALLOW storage:buckets:read WHERE storage:bucket-name = "audit_logs_365d";
ALLOW storage:logs:read, storage:metrics:read, storage:spans:read;
```

Verifying Bucket Access

Best Practices

Practice	Rationale
Use consistent bucket naming	Enables pattern-based policies
Assign policies to groups, not users	Easier management, consistent access

Practice	Rationale
Document all policies	Audit trail and governance
Test policies with sample users	Prevent access issues
Use STARTSWITH for team prefixes	Future-proof for new buckets
Combine with record-level for scale	Bucket limits (80) may not suffice

When Bucket-Level Isn't Enough

Bucket-level access has limitations:

Limitation	Alternative
80 bucket limit	Use security context for finer granularity
Shared data needs	Use record-level permissions
Dynamic team membership	Use security context mapped to tags
Field-level masking	Use field-level permissions

See **ORGNZ-06** and **ORGNZ-07** for advanced permission patterns.

Next Steps

Continue with the ORGNZ series:

- **ORGNZ-06**: Security Context

References

- [Permissions in Grail](#)
- [IAM policy reference](#)

This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.

DQL: Verifying Bucket Access

Confirm you can read from the expected buckets:

```
// Verify bucket access – check data distribution across buckets you can read
fetch logs, from:-1h
| summarize count = count(), by:{dt.system.bucket}
| sort count desc
```